

Assessing the burden of congenital toxoplasmosis in Burundi, 2020

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ABSTRACT

Congenital toxoplasmosis (CT) due to *Toxoplasma gondii*, an apicomplexan parasite, leads to significant sequelae in children, foetal losses and neonatal deaths worldwide. This study aimed to assess the burden of CT in Burundi for the year 2020. We used epidemiological and economic data collected in major hospitals in Burundi, the Ministry of Public Health in Burundi and international peer-reviewed literature to estimate the disability-adjusted life years (DALYs) and economic costs spent on prenatal consultations, diagnosis and treatment of toxoplasmosis for pregnant women. In Burundi, the incidence of CT in 2020 was 2.7 cases per 1000 children (95 % UI: 1.8–3.7) and 1180 children (95 % UI: 797–1633) were born with CT. The health burden due to CT in 2020 was estimated at 7283 DALYs (95 % UI: 4469–10,920) when including and 3671 DALYs (95 % UI: 2286–5547) when excluding foetal losses, respectively. Approximately 550,000 USD (95 % UI: 222,027–1,088,331) were economic losses due to CT in 2020, with 690 USD (95 % UI: 375–1002) per pregnant woman receiving treatment until delivery and 31 USD (95 % UI: 10–70) for an untreated one. This study demonstrates that CT is a significant public health concern in Burundi. However, due to data gaps, uncertainties in DALY and economic cost estimates remain. Control measures, including proper meat cooking, improved hygiene and sanitation practices, better management of cats and their faeces, routine diagnostic check-ups for pregnant women, and timely treatment for infected pregnant women, are crucial to reduce the health and economic burden of toxoplasmosis in Burundi.

1. Introduction

Toxoplasmosis is a cosmopolitan zoonosis caused by an apicomplexan parasite, *Toxoplasma gondii* (*T. gondii*), which affects humans and animals (Dubey et al., 2021). The parasite's life cycle is complex, with cats and other felids acting as the definitive hosts, and humans and all other warm-blooded animals acting as intermediate hosts (Dubey, 2009). *Toxoplasma gondii* is ranked among major foodborne parasites with high health impacts in humans (Robertson et al., 2024; van der Giessen et al., 2021). Humans get infected with *T. gondii* in several ways, including the ingestion of food or water contaminated with oocysts from cat faeces, ingestion of undercooked meat containing *T. gondii* cysts, transplacental transmission during pregnancy, blood transfusion, organ transplantation and accidental inoculation of *T. gondii* tachyzoites

(Dubey, 2009; Stelzer et al., 2019). Most people infected with *T. gondii* do not show clinical signs, except for immunocompromised individuals who can develop encephalitis and children infected through maternal-foetal transmission (Bollani et al., 2022; Weiss and Dubey, 2009). Studies have shown that maternal-foetal transmission is low (6–15 %) during the first trimester of pregnancy, moderate (30–40 %) during the second trimester, and very high (over 70 %) during the third trimester (Bollani et al., 2022; Dunn et al., 1999; Rzaq et al., 2023). Maternal-foetal transmission between 10 and 24 weeks of gestation is associated with severe clinical signs of congenital toxoplasmosis (CT), such as abortion, stillbirth, or sequelae after birth, while transmission between 26 and 40 weeks is associated with subclinical signs that often become evident later in life (Bollani et al., 2022; Rzaq et al., 2023; Weiss and Dubey, 2009). Children with CT can develop clinical signs including

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chorioretinitis, strabismus, blindness, epilepsy, psychomotor or mental retardation, hydrocephalus, encephalitis, microcephaly, intracranial calcification, anaemia, jaundice, rash, petechiae due to thrombocytopenia, pneumonitis, hypothermia, diarrhoea, and non-specific illness (Bollani et al., 2022; Dubey et al., 2021; Weiss and Dubey, 2009). Despite the numerous sequelae, only chorioretinitis, hydrocephalus, intracranial calcification, and central nervous system (CNS) abnormalities are considered when defining the burden of CT in children (Bollani et al., 2022; Havelaar et al., 2007; Rzaq et al., 2023; Weiss and Dubey, 2009).

To better monitor toxoplasmosis in pregnant women, *Toxoplasma*-specific immunoglobulin M (IgM) and immunoglobulin G (IgG) tests are used during pregnancy (Bollani et al., 2022; Gras et al., 2004; Villard et al., 2016). The global seroprevalence of toxoplasmosis in pregnant women based on IgG was estimated at 33.8 %, with the highest rates observed in South America at 56.2 % and Africa at 48.7 %, and the lowest in the Western Pacific region at 11.8 % (Rostami et al., 2020). A study carried out in African countries reported a 42.9 % seroprevalence of toxoplasmosis in pregnant women (Mulu Gelaw et al., 2024). The detection of specific IgG antibodies, combined with the absence of IgM, indicates a past *Toxoplasma* infection (Bollani et al., 2022; Villard et al., 2016). Only acute toxoplasmosis in women can lead to maternal-foetal transmission when it occurs during pregnancy or shortly before conception (Rostami et al., 2019). The diagnosis of acute toxoplasmosis during pregnancy is based on the detection of specific IgM antibodies, low-avidity anti-*Toxoplasma* IgG antibodies, and seroconversion from IgG-negative to IgG-positive status in cases where sequential testing is performed during pregnancy (Rostami et al., 2019; Villard et al., 2016). The global prevalence of acute toxoplasmosis in pregnant women was estimated at 1.1 %, with 1.6 % reported in Africa (Rostami et al., 2019). Furthermore, comprehensive data on the detection of toxoplasmosis in pregnant women based on IgM or IgG are lacking for most African countries, including Burundi (Mulu Gelaw et al., 2024; Rostami et al., 2019), creating a gap in the follow-up of pregnant women infected with toxoplasmosis and leaving them untreated. In some Western European countries, monthly or bimonthly screening for toxoplasmosis in pregnant women was implemented to enable early treatment, thereby preventing the sequelae of CT in children (Rostami et al., 2019). A low incidence of CT, estimated at 5.5 cases per 100,000 live births, was recently reported in Europe (ECDC, 2024). However, the updated incidence of CT in African countries remains unknown, although it is expected to be high due to risky practices associated with poor sanitation and culinary habits (Mulu Gelaw et al., 2024; Rostami et al., 2019).

In Burundi, inadequate practices related to meat handling and preparation, consumption of undercooked meat, use of unsafe water, eating raw vegetables, and inappropriate washing of vegetables and fruits have been reported as major risk factors for human toxoplasmosis, which is particularly severe in pregnant women due to the risk of transplacental transmission (Minani et al., 2025a). A burden of disease study is a powerful tool for providing evidence-based results to policy makers, informing them about the magnitude of the disease in a country and helping them allocate resources effectively (Torgerson et al., 2015). Studies on the burden of CT have been mostly conducted in European countries; however, they are lacking for the African continent (Havelaar et al., 2007; Nissen et al., 2017; Smit et al., 2017). The present study aimed to estimate the burden of CT in Burundi for the year 2020, using data from major hospitals in Burundi, the Ministry of Public Health in Burundi, and international peer-reviewed literature.

2. Methods

2.1. Study area and population

Burundi is a small landlocked country in East Africa, covering an area of 27,834 km². The total population in 2020 was estimated at 11,215,758 people, including 5,529,432 males and 5,686,146 females

(ISTEEBU, 2013) (Table 1). In Burundi, the median age at first birth among women is 21.5 years, and the average number of children per woman is 5.5 (MSPLS, 2018). Neonatal deaths are estimated at 23 per 1000 live births (MSPLS, 2021). In 2020, the health system structure comprised 149 hospitals (52 public, 69 private, and 28 denominational) and 1182 health centres (658 public, 380 private, and 144 denominational) (MSPLS, 2021). Since 2006, almost all preventive services and curative care for children under five years old, as well as for pregnant women and women giving birth, have been provided free of charge at public health facilities (MSPLS, 2021). In terms of social protection, 58.5 % of the population is covered by mutual health insurance, which includes 23 % with Medical Assistance Cards (CAM), 22 % with free coverage, 6.5 % with Mutual Civil Service insurance, and 7 % with other health insurance mechanisms (MSPLS, 2021). Moreover, 87 % of the population has access to health facilities within a radius of less than 5 km (MSPLS, 2021; OMS, 2017). However, access to quality healthcare in rural areas remains a significant issue, as the best hospitals with advanced diagnostic equipment and skilled doctors are concentrated in Bujumbura city (MSPLS, 2021; OMS, 2017).

2.2. Health and economic burden estimation of CT

The health and economic burden of CT was assessed in accordance with the Standardised Reporting of Burden of Disease studies (STROBOD) statement (Devleesschauwer et al., 2024). A retrospective descriptive cross-sectional study was conducted in Burundi for the year 2020, utilising data from hospital registers, the Ministry of Public Health, and published literature to estimate the health burden in terms of disability-adjusted life years (DALYs) and the economic burden, including costs of medical consultations, diagnosis, and medication.

2.2.1. Health burden estimation parameters for CT

2.2.1.1. Data parameters. To estimate the burden, a decision tree for CT was built (Fig. 1). The health burden in 2020 in Burundi was assessed using disability-adjusted life years (DALYs), which is the sum of years lived with disability (YLD) and years of life lost due to premature mortality (YLL) (Devleesschauwer et al., 2014b). To estimate YLDs and YLLs, epidemiological data were collected from various sources, including hospitals, the Ministry of Public Health, and international peer-reviewed literature. Data on overall live births, stillbirths, and abortions were collected from the Ministry of Public Health for the year 2020 (Table 2). Due to the lack of seroprevalence data for toxoplasmosis in pregnant women and the incidence of CT in newborns, several hospitals were visited between December 2022 and April 2023 to collect this data. These hospitals included Centre Hospitalo Universitaire de Kamenge (CHUK), Hôpital Militaire de Kamenge (HMK), Kayanza Hospital, Ngozi Hospital (public hospitals), and Kira Hospital (a private hospital). These hospitals were selected based on their location and national and provincial rankings. Due to the high concentration of hospitals in Bujumbura city, CHUK and HMK were chosen because they are among Burundi's four national reference hospitals, while Kira Hospital is one of the highly regarded private hospitals. In the countryside, provincial hospitals such as Kayanza and Ngozi were included in the study because these provinces are densely populated. During these

Table 1
Human population by age and sex categories in 2020.

Age (years)	Male	Female	Total
0–4	893,872	888,554	1782,426
5–14	1486,390	1514,133	3000,523
15–44	2459,411	2568,012	5027,423
45–59	470,882	481,213	952,095
60+	218,877	234,234	453,111
All ages	5,529,432	5,686,146	11,215,758

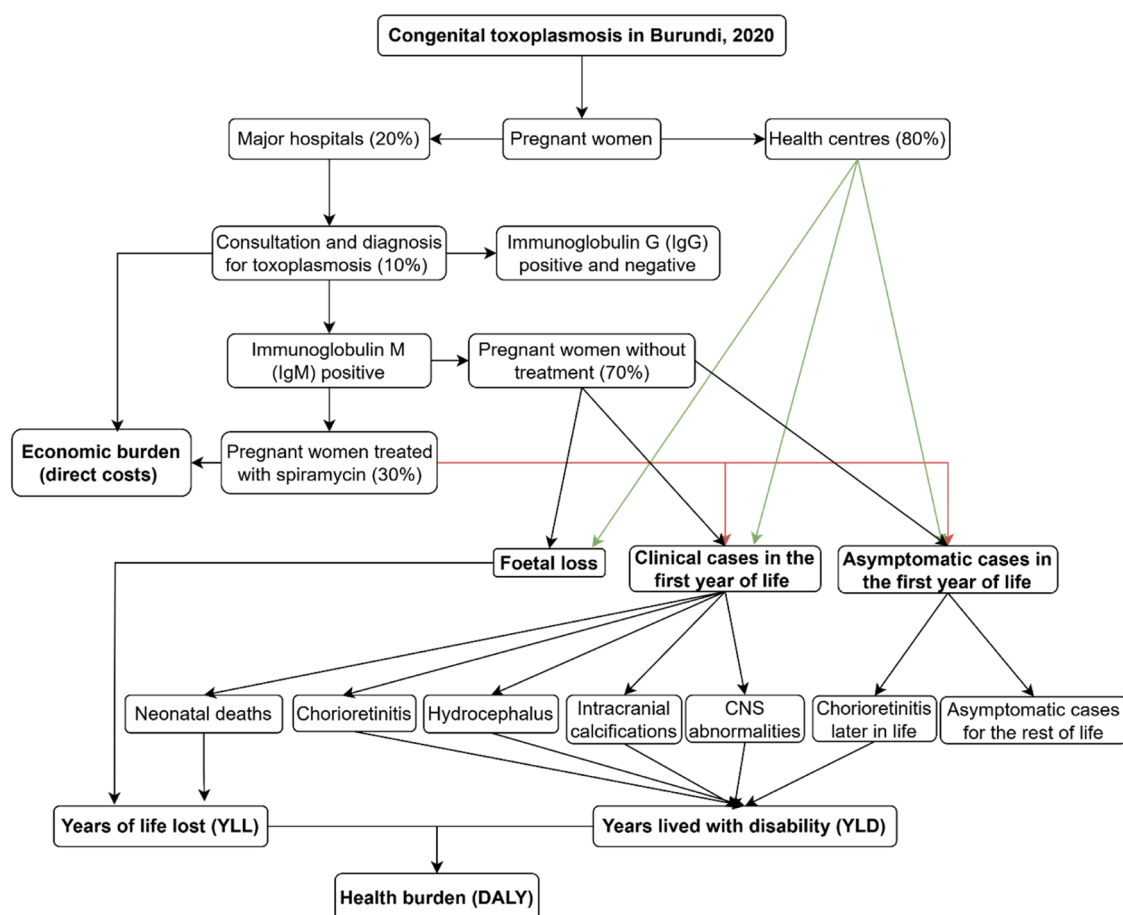


Fig. 1. Decision tree to estimate the health and economic burden in Burundi in 2020, adapted from Nissen et al. (2017) and Devleeschauwer et al. (2017) (CNS: Central nervous system).

Table 2

Data collected in the Ministry of Public Health to estimate the health burden in Burundi, 2020.

Notation	Data	Distribution (values)	Sources
Lb	Number of live births in 2020	Fixed (437,685)	MPH
Sb	Stillbirths after gestational week 22 in 2020	Fixed (7505)	MPH
Sa	Number of reported spontaneous abortions occurring up to 22 weeks of gestation	Fixed (11,616)	MPH
TP	Total number of pregnancies (mothers) in 2020	Lb+Sb+Sa	–

MPH: Ministry of Public Health.

visits, laboratory diagnoses of toxoplasmosis in pregnant women for the year 2020 were retrieved from the hospital archives, focusing on the number of cases tested for *T. gondii*-specific IgG or IgM antibodies (Table 3) (Villard et al., 2016). Additionally, short interviews were conducted with the staff responsible for gynaecological and obstetric services at each hospital to collect information on prenatal consultations, toxoplasmosis diagnosis results, and the treatment of toxoplasmosis cases.

2.2.1.2. Methodological procedures. To estimate the incidence of CT, we used methods similar to those described by Havelaar et al. (2007) and Kortbeek et al. (2009). For the analysis, we used the proportion of seronegative pregnant women, the probability of pregnant women getting infected with *T. gondii* based on the detection of IgM antibodies, and

Table 3

Data collected in hospitals to estimate the health burden in Burundi, 2020.

Data	Total pregnant women	Positive IgG	Negative IgG	Positive for IgM
CHUK	1227	769	458	5
HMK	2358	232	2126	3
Ngozi hospital	144	83	61	9
Kayanza hospital	177	39	138	8
Kira hospital	424	75	349	5
Total	4330	1198	3132	30 / 3132
%		27.67 (26.33–28.99)	72.33 (70.9–73.6)	0.96 (0.65–1.33)

CHUK: Centre Hospitalo Universitaire de Kamenge; HMK: Hôpital Militaire de Kamenge; IgG: Immunoglobulin G; IgM: Immunoglobulin M.

the probability of maternal seroconversion to the foetus (Table 4). Maternal seroconversion to the foetus has been reported to range from 25 % to 29 % (Rzaq et al., 2023). We considered a 28.1 % proportion of maternal seroconversion to the foetus in the analyses (Torgerson and Mastroiacovo, 2013). The number of neonatal deaths and live births infected with CT was calculated by multiplying the total number of pregnancies by the incidence of CT per 1000 live births.

YLDs were calculated by multiplying the number of incident cases based on the sequelae of CT by the duration of each sequela and the associated disability weights (DW). The same sequelae proportions reported by Havelaar et al. (2007) were applied to children developing

Table 4

Epidemiological parameters used to estimate the health burden in Burundi, 2020.

Notation	Epidemiological parameters	Distribution (values)	Sources
PSeroneg	Proportion of seronegative pregnant women	Beta (3132,1198)	Hospital data
PSerop	Probability of pregnant women with IgM-positive	Beta (30, 3102)	Hospital data
NSeroconv	Number of seroconverting pregnant women	$TP \times PSeroneg \times PSerop$	–
PMF	Probability of seroconversion from mother to foetus	Fixed (0.281)	(Torgerson and Mastroiacovo, 2013)
PFL	Probability of foetal loss in seroconverting mothers	Uniform (0.017, 0.04)	(Binquet et al., 2003; Havelaar et al., 2007)
NFL	Number of foetal losses due to CT >22 weeks of gestation	$NSeroconv \times PFL \times ((40-22)/40)$	–
IncCT	Incidence of CT per 3132 live births	$NSeroconv \times PSerop \times PMF$	–
IncCT-bi	Incidence of CT per 1000 live births in Burundi	$(IncCT \times 1000)/3132$	–
Cases CT	Total number of postnatal with CT	$(TP \times IncCT-bi)/1000$	–
Pch1	Probability of children with chorioretinitis at birth	Beta (141, 907)	(Havelaar et al., 2007)
Pch2	Probability of children with chorioretinitis later in life	Fixed (0.16)	(Torgerson and Mastroiacovo, 2013)
Phydro	Probability of children with hydrocephalus	Beta (16,840)	(Havelaar et al., 2007)
Picc	Probability of children with intracranial calcifications	Beta (88,749)	(Havelaar et al., 2007)
PCNS	Probability of children with CNS abnormalities	Beta (3, 102)	(Havelaar et al., 2007)
Pnd	Probability of neonatal death	Beta (9, 1202)	(Havelaar et al., 2007)
DWCh1	Disability weight for children with chorioretinitis at birth	Fixed (0.089)	(WHO, 2020)
DWCh2	Disability weight for children with chorioretinitis later in life	Fixed (0.089)	(WHO, 2020)
DWH	Disability weight for children with hydrocephalus	Fixed (0.36)	(Torgerson and Mastroiacovo, 2013; WHO, 2015)
DWICC	Disability weight for children with intracranial calcifications	Fixed (0.01)	(Torgerson and Mastroiacovo, 2013; WHO, 2015)
DWCNS	Disability Weight for children with CNS abnormalities	Fixed (0.36)	(Torgerson and Mastroiacovo, 2013; WHO, 2015)
DWND	Disability weight for neonatal death	Fixed (1)	(Torgerson and Mastroiacovo, 2013)
DWFL	Disability weight for foetal loss	Fixed (1)	(Torgerson and Mastroiacovo, 2013)

chorioretinitis at birth, hydrocephalus, CNS abnormalities and intracranial calcifications. In addition, we considered 16 % for those who may develop chorioretinitis later in life, after 10 years (Torgerson and Mastroiacovo, 2013). Based on the life expectancy at birth in Burundi (MSPLS, 2023), we assumed that all sequelae were lifelong, reaching up to 59.7 years. For children developing chorioretinitis later in life (after 10 years), we assumed that the duration of the sequelae could reach up to 49.7 years (Havelaar et al., 2007; WHO, 2015). For the following health states: hydrocephalus, intracranial calcification and CNS abnormalities, we used the same DW as Torgerson and Mastroiacovo (2013) and WHO (2015). For chorioretinitis, the updated DW based on distance vision (moderate impairment) from WHO (2020) was used (Table 4).

YLLs were estimated based on the number of foetal losses, neonatal deaths and the standard life expectancy of 88.87 years used in the GBD 2019 (Global Burden of Disease, 2019) (Table 4). To obtain the number of foetal losses, we assumed that the mean probability of foetal loss ranges from 1.7 % to 4 % in seroconverting mothers (Binquet et al., 2003; Havelaar et al., 2007) and that foetal loss could occur after 22 of the 40-week gestation period (Nissen et al., 2017). We also used the same proportion of neonatal deaths as reported by Havelaar et al. (2007) and Torgerson and Mastroiacovo (2013). Although 87 % of the population in Burundi can have access to health facilities within a radius of less than five kilometres (MSPLS, 2021), most health centres serving large populations do not diagnose toxoplasmosis. Toxoplasmosis is diagnosed only in major hospitals across the country (personal communications with medical doctors). We estimated the health impact of toxoplasmosis in pregnant women, considering both those receiving treatment to prevent maternal transmission to the foetus and those not receiving treatment. Spiramycin is widely used in hospitals for pregnant women who are detected with IgM-positive antibodies (Binquet et al., 2019; Bollani et al., 2022). However, based on information obtained during hospital visits, field experience and the low standard of living of less than 2 USD per day in Burundi (The World Bank, 2020), we assumed that only 3 % of pregnant women in Burundi who tested positive for toxoplasmosis received medication. Spiramycin has been shown to reduce maternal transmission of toxoplasmosis to the foetus by 50 % (Binquet et al., 2019). Thus, a 50 % reduction in toxoplasmosis sequelae in children from treated pregnant women was considered, except for foetal loss. In the overall estimates, two scenarios were considered: one including foetal loss and another excluding foetal loss, as performed by Nissen et al. (2017) and Torgerson and Mastroiacovo (2013), assuming that the loss of a foetus by a mother cannot cause any sequelae for the mother because she may soon become pregnant.

2.2.2. Cost estimation parameters for CT

The economic impact of toxoplasmosis was estimated only for pregnant women visiting health facilities for prenatal consultations, as well as for the diagnosis and treatment of toxoplasmosis (Table 5). Since toxoplasmosis is not diagnosed due to a lack of diagnostic tools in most health centres, which account for approximately 80 % of health facilities in the country (MSPLS, 2021), most pregnant women are not screened for the disease. In addition, in some major hospitals capable of diagnosing toxoplasmosis, reagent shortages lasting one to three months were observed when collecting data on pregnant women diagnosed with toxoplasmosis. Given these limitations, we assumed that only 10 % of pregnant women could be tested for toxoplasmosis at least once during pregnancy in cases with negative or IgG-positive results (Fig. 1). We estimated that 10 % of the pregnant women who seroconverted during pregnancy had two to four visits for medical consultations and toxoplasmosis tests, with an average of three visits (personal communications with medical doctors). We also assumed that only 30 % of the 10 % of seroconverting pregnant women received spiramycin treatment and adhered to it until delivery (personal communications with medical doctors). Since maternal-foetal transmission is high during the second and third trimesters of pregnancy (Binquet et al., 2019; Bollani et al., 2022; Rzaq et al., 2023), we considered that the duration of spiramycin

Table 5

Epidemiological parameters used to estimate the economic burden.

Parameter	Distribution	Values	Sources
Total number of pregnancies (mothers) in 2020	Fixed	456,806	MPH data
Proportion of pregnant women visiting hospitals for toxoplasmosis consultation	Fixed	0.1	Personal communication
Proportion of seronegative pregnant women	Beta (3132,1198)	0.723 (0.709–0.736)	Hospital data
Probability of pregnant women with IgM-positive	Beta (30, 3102)	0.00959 (0.00648–0.0133)	Hospital data
Number of hospital visits for seroconverting pregnant women	Uniform (2,4)	3 (2.05–3.95)	Hospital data
Cost of medical consultation (BIF)	Gamma (3.31, 6.61e-04)	5007 (1145–11,660)	Hospital data
Cost of toxoplasmosis diagnosis (BIF)	Gamma (2.94, 1.96e-04)	15,010 (3004–36,520)	Hospital data
Cost of spiramycin treatment (90 pills per month) in (BIF)	Uniform (270,000,360,000)	315,000 (272,300–357,780)	Hospital data
Duration of spiramycin treatment (months)	Uniform (2,6)	4 (2.1–5.9)	Personal communication

treatment could range between two and six months, with an average of four months (personal communication with medical doctors). Due to insufficient data, the analysis did not include costs for pregnant women hospitalised for toxoplasmosis-related issues, children with toxoplasmosis sequelae, or indirect costs related to employment or disability.

2.3. Data analysis

All the analyses were performed using R software version 4.3.3 (R Core Team, 2024). The 95 % uncertainty interval (UI) for each parameter, including the median (50 % quantile) and the 2.5 % and 97.5 % quantiles, was calculated using Monte Carlo simulations with 100,000 iterations. Various probability distributions were used depending on the type of variable. The lowest and highest values were fixed to estimate the mean for a uniform distribution. The uniform distribution was used to model the probability of foetal loss in seroconverting mothers, with minimum and maximum values of 1.7 % and 4 %, respectively; the number of hospital visits, with a minimum of 2 and a maximum of 4; and the duration of treatment, with a minimum of 2 months and a maximum of 6 months. A Gamma distribution enabled us to model the costs associated with consultation and diagnostic services since their prices vary from one hospital to another. A shape parameter (α) and a rate parameter (β) for the Gamma distribution were obtained using an online Gamma distribution calculator (AgriMetSoft, 2019), based on the various costs associated with medical consultation and diagnostic tests in different hospitals. For example, the costs of medical consultations ranged from 3000 to 6000 BIF (Burundi francs) in public hospitals and from 10,000 to 20,000 BIF in private hospitals. Since a high proportion

of women visited public hospitals compared to private hospitals, the estimated shape ($\alpha = 3.31$) and rate ($\beta = 0.000661$) parameters resulted in a mean cost of 5007 BIF. Moreover, a Beta distribution enabled us to model adjustments on diagnostic uncertainties and some proportions of CT sequelae in children. Two positive shape parameters, α and β , were set for the Beta distribution (Zhang et al., 2025), for example, with $\alpha = 30$ and $\beta = 3132-30$, to estimate the probability of pregnant women with IgM-positive. Only the scenario of DALYs [0;0] without age weighting ($K = 0$) or time discounting ($r = 0$) was reported as performed in most GBD studies (Devleeschauwer et al., 2024, 2014a; Nissen et al., 2017; Torgerson and Mastroiacovo, 2013). The economic costs were estimated using a 2020 exchange rate of 1915 BIF per 1 USD (United States dollar). Scripts containing all health and economic burden estimates and sensitivity analyses are available online at https://github.com/MINANI-Salvator/Burden-Toxo_Burundi.

2.4. Ethics statement

The research protocol was approved by the National Ethics Committee in Burundi (CNE/25/2022) and the Institutional Review Board of the Institute of Tropical Medicine (ITM) in Belgium (IRB/RR/AC/091 Ref 1595/22). Permissions required to collect data in different hospitals and the Ministry of Public Health were obtained from the department heads or directors. Anonymised data from hospital registers were recorded. Oral consent was obtained from the medical staff when asked for some information about toxoplasmosis, such as consultation with pregnant women, diagnostics and treatment.

3. Results

3.1. Incidence and number of cases of CT

The incidence of CT was 2.7 per 1000 children (95 % UI: 1.8–3.7). We estimated that of the 456,806 pregnancies in Burundi in 2020, 3169 women (95 % UI: 2142–4392) were infected with toxoplasmosis, resulting in 41 foetal losses (95 % UI: 21–67). In addition, 1180 children (95 % UI: 797–1633) were born with CT, including 1144 children from untreated pregnant women (95 % UI: 773–1584) and 36 children (95 % UI: 24–49) from pregnant women treated with spiramycin (Table 6). Based on the sequelae of CT, of the 1180 children, 519 (95 % UI: 347–729) had clinical signs at birth or later in life, 9 experienced neonatal death (95 % UI: 4–16), and 652 (95 % UI: 436–911) were

Table 6

Incidence and cases of CT.

Estimate	Values/estimate
Incidence of CT per 1000 births	2.7 (1.8–3.7)
Total cases of CT	1180 (797–1633)
Children with CT from untreated women	1144 (773–1584)
Children with CT from treated women	36 (24–49)
Total children with chorioretinitis at birth	156 (102–223)
Children with chorioretinitis at birth from untreated women	154 (101–220)
Children with chorioretinitis at birth from treated women	2 (1–3)
Total children with chorioretinitis later in life	186 (126–257)
Children with chorioretinitis later in life from untreated women	183 (124–253)
Children with chorioretinitis later in life from treated women	3 (2–4)
Total children with hydrocephalus	22 (11–37)
Children with hydrocephalus from untreated women	21 (11–36)
Children with hydrocephalus from treated women	1 (0–1)
Total children with intracranial calcification	122 (78–177)
Children with intracranial calcification from untreated women	120 (77–175)
Children with intracranial calcification from treated women	2 (1–3)
Total children with CNS abnormalities	33 (7–82)
Children with CNS abnormalities from untreated women	32 (7–81)
Children with CNS abnormalities from treated women	1 (0–1)
Total neonatal deaths	9 (4–16)
Neonatal deaths from untreated women	8.5 (4–16)
Neonatal deaths from treated women	0.1 (0.05–0.25)
Total number of foetal losses	41 (21–67)

considered asymptomatic for the rest of their lives (Table 6).

3.2. Health burden estimation

The health burden of CT in Burundi for the year 2020 was estimated at 7283 DALYs (95 % UI: 4469–10,920), with 2904 YLDs (95 % UI: 1802–4456) and 4379 YLLs (95 % UI: 2420–6993) (Table 7). From the total DALYs, foetal loss contributed to 3612 DALYs (95 % UI: 1888–5942), representing 49.1 % of the total. The estimated number of DALYs per 100,000 person-years was 64.9 (95 % UI: 39.9–97.4) and the DALYs per incident case were 6.2 (95 % UI: 4.7–7.7). When excluding foetal loss from the overall DALY estimates, the health burden of CT was reduced by half, resulting in 3671 DALYs (95 % UI: 2286–5547), of which 79.2 % (95 % UI: 66.9–89.6) were attributable to YLDs and 20.8 % (95 % UI: 10.4–33.1) to YLLs (neonatal deaths). The estimated number of DALYs per 100,000 person-years was 32.7 (95 % UI: 20.4–49.5) and the DALYs per incident case were 3.1 (95 % UI: 2.4–4.1).

3.3. Economic burden estimation

The economic losses due to CT were estimated at 549,949 USD (95 % UI: 222,027–1,088,331) (Table 8). Costs per case associated with toxoplasmosis were 690 USD (95 % UI: 375–1002) for cases receiving treatment until delivery and 31 USD (95 % UI: 10–70) for those not receiving any treatment.

3.4. Sensitivity analyses

The proportion of IgM-positive, the proportion of neonatal deaths, and the proportion of CNS abnormalities contributed most to the overall health burden (Fig. 2 and 3). Diagnostic cost, medical consultation cost and the duration of treatment contributed most to the overall economic

Table 7
Estimated health burden associated with CT in Burundi, 2020.

Estimate	Mean (95 % UI)	Contribution (%) in DALY	Contribution (%) in DALY*
YLD			
Chorioretinitis at birth	831 (544–1186)	11.6 (8.7–15.5)	23.0 (16.9–29.6)
Chorioretinitis later in life	822 (556–1138)	11.5 (9–14.9)	22.8 (17.2–28.7)
Hydrocephalus	466 (236–796)	6.5 (3.6–10.4)	12.8 (7.3–19.7)
Intracranial calcifications	73 (47–106)	1.0 (0.7–1.4)	2.0 (1.4–2.7)
CNS abnormalities	712 (141–1767)	9.7 (2.2–21.2)	18.6 (4.8–36.8)
Total	2904 (1802–4456)	40.3 (30.2–52.7)	79.2 (66.9–89.6)
YLL			
Foetal loss	3612 (1888–5942)	49.1 (35.9–60.7)	–
Neonatal deaths	767 (317–1454)	10.6 (4.9–18.3)	20.8 (10.4–33.1)
Total	4379 (2420–6993)	59.7 (47.3–69.8)	20.8 (10.4–33.1)
DALYs			
Total (YLD+YLL)	7283 (4469–10,920)	100	–
Total DALYs*	3671 (2286–5547)		100
DALYs per 100,000 person-years	64.9 (39.9–97.4)		
DALYs* per 100,000 person-years	32.7 (20.4–49.5)		
DALYs per incident case	6.2 (4.7–7.7)		
DALYs* per incident case	3.1 (2.4–4.1)		

* Excluding foetal loss in the estimates; YLD: Year lived with disability; YLL: Year of life lost; CNS: Central nervous system; DALYs: Disability-adjusted life years; UI: Uncertainty interval.

Table 8

Estimated economic costs associated with CT.

Type of costs	Mean of USD (95 % UI)	% of total costs
Consultation	121,717 (27,941–284,851)	24.1 (5.3–54.5)
Diagnostic	365,638 (73,523–887,249)	62.8 (26.9–87.1)
Medicine	62,594 (28,800–108,374)	13.1 (4.2–30.3)
Total cost	549,949 (222,027–1,088,331)	100
Cost per treated case	690 (375–1002)	–
Cost per untreated case	31 (10–70)	–

USD: United States dollars; UI: Uncertainty interval.

losses (Fig. 4).

4. Discussion

The current study estimated the burden of CT in Burundi for the year 2020. The burden was estimated at 7283 DALYs with foetal loss included and 3671 DALYs without foetal loss. To our knowledge, this study is the first to assess the burden of CT for an African country. The health burden of this disease in Burundi was much higher than the burden reported in European countries, where 123 DALYs were estimated in Denmark in 2014, 188 DALYs in Belgium in 2013, and 620 and 2303 DALYs in the Netherlands in 2007 and 2009, respectively (Havelaar et al., 2007; Kortbeek et al., 2009; Nissen et al., 2017; Smit et al., 2017). The high burden in Burundi could be attributed to the high incidence of CT, with a rate of 2.7 per 1000 live-born children, compared to 5.5 per 10,000 in Belgium and 1.9 per 10,000 in Denmark, respectively (Nissen et al., 2017; Smit et al., 2017). Despite the high life expectancy used to estimate YLDs in Belgium (80.25 years), Denmark (81 years), and the Netherlands (79 years), their YLDs remained lower than the estimated YLDs in Burundi, where a life expectancy of 59.7 years was used. The reported burden of CT in Nepal (9255 DALYs) was almost similar to our results, although the DALYs per 100,000 person-years estimated at 35.2 were almost half of those found in Burundi (Devleesschauwer et al., 2014a). Compared to major infectious diseases in Burundi such as malaria, tuberculosis, HIV/AIDS, fungal infections and *Klebsiella pneumoniae* infections, the burden of CT appears to be low. In 2019, the estimated health burden for malaria in Burundi was 714,000 DALYs, followed by tuberculosis at 541,000 DALYs, HIV/AIDS at 132,000 DALYs, fungal infections at 125,000 DALYs and *Klebsiella pneumoniae* infections at 120,000 DALYs (Naghavi et al., 2024). In addition, *Taenia solium* cysticercosis, a neglected tropical disease, led to 14,603 DALYs in Burundi in 2020, approximately twice the burden of CT (Minani et al., 2022).

Considering foetal loss in the DALY estimates, 63.4 % of the DALYs were attributed to foetal loss in Denmark, a figure higher than that of Burundi (49.1 %) or of the Netherlands (36.6 %) (Kortbeek et al., 2009; Nissen et al., 2017). Although the foetal loss was excluded from the DALY estimates in Belgium, it represented a proportion (52.3 %) almost similar to that of Burundi (Smit et al., 2017). There is no established methodology for estimating foetal loss in DALY calculations due to the complexity of multiple contributing factors. It is up to researchers to include or exclude it in the overall DALY estimates based on the country context. In some European countries, there is a good follow-up of pregnant women, which could prevent or decrease the rate of foetal losses; hence, the incidence of CT is very low, estimated at 5.5 cases per 100,000 live births (ECDC, 2024). However, from a control intervention perspective in Burundi, it is important to consider the impact of foetal loss in the DALY estimates, given the high incidence of CT (2.7 cases per 1000 live births), to strengthen diagnostic capacity and the use of spiramycin treatment. In addition, its exclusion from the DALY estimates may lead to biased results, as toxoplasmosis has been reported to be more severe when contracted in the first and second trimesters of pregnancy than in the third (Bollani et al., 2022; Dunn et al., 1999; Rzaq et al., 2023). Given the severe clinical signs and sequelae after birth of CT, it is crucial to treat infected pregnant women early to prevent

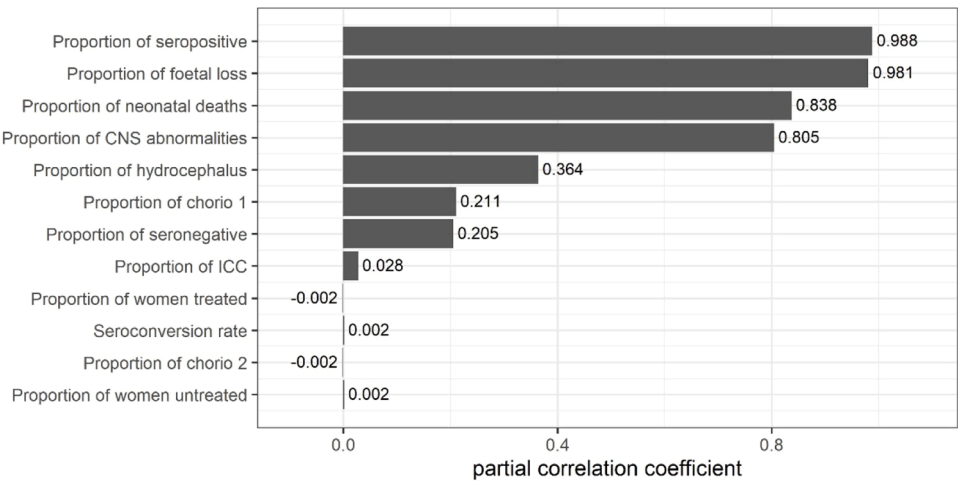


Fig. 2. Partial correlation coefficients of parameters that influence the estimated DALYs (CNS: Central nervous system, chorio1: chorioretinitis at birth, chorio2: chorioretinitis later in life, ICC: Intracranial calcifications).

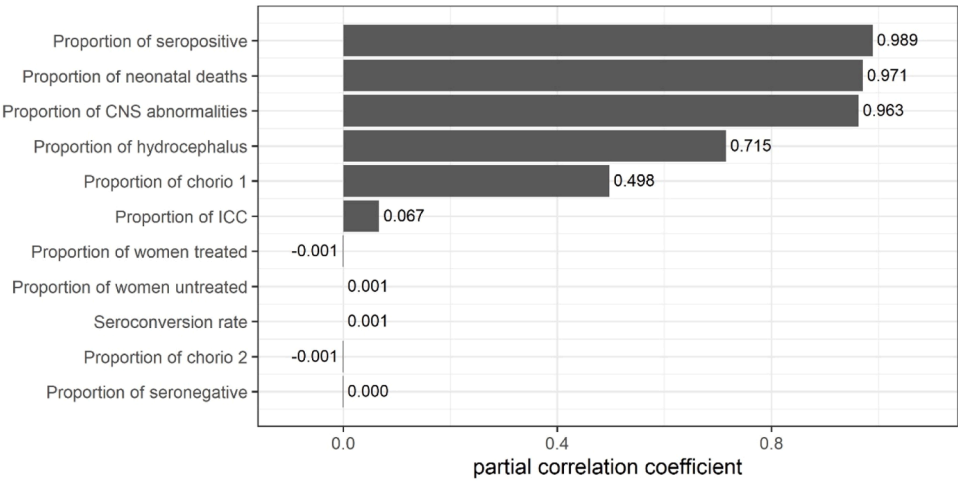


Fig. 3. Partial correlation coefficients of parameters that influence the estimated DALYs without foetal loss (CNS: Central nervous system, chorio1: chorioretinitis at birth, chorio2: chorioretinitis later in life, ICC: Intracranial calcifications).

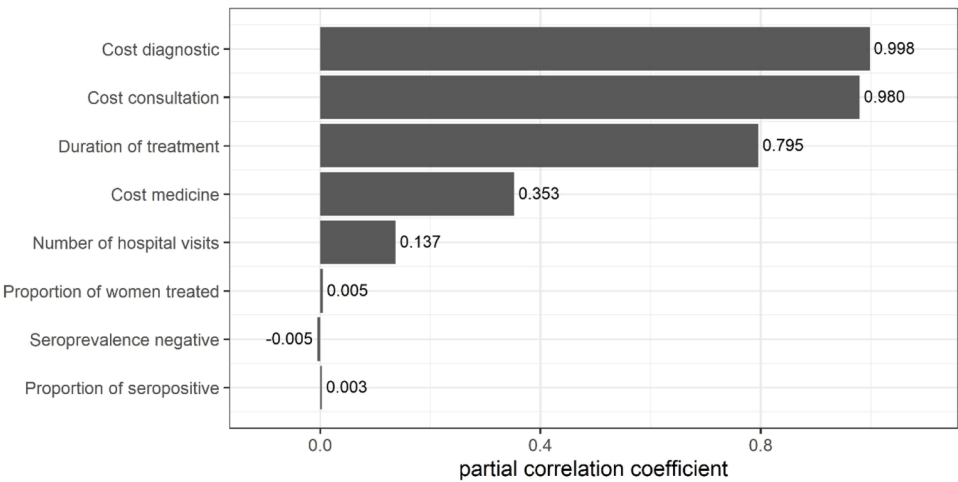


Fig. 4. Partial correlation coefficients of parameters that influence the estimated costs.

maternal-foetal transmission. In this study, we assumed that only a small proportion of infected pregnant women took spiramycin during pregnancy, as the diagnostic capacity is only available in major hospitals. The health burden in this study was higher than that reported for Burundi in the global burden of CT in 2013 (6843 DALYs) (Torgerson and Mastroiacovo, 2013). Although spiramycin is recommended for infected pregnant women, its effectiveness in preventing maternal-foetal transmission is only 50 % (Binqet et al., 2019). In addition, Havelaar et al. (2007) found no clear evidence regarding the efficacy of prenatal and postnatal treatments in reducing clinical signs of maternal-foetal transmission and in affected children, respectively (Havelaar et al., 2007). In Burundi, data on the follow-up of pregnant women taking medication and the postnatal stage were missing, which could lead to either an overestimation or underestimation of the results.

The economic impact of CT was estimated at 549,949 USD, of which 690 USD was spent on consultation, diagnosis, and treatment for one pregnant woman until delivery. This economic burden appears high and unsupportable for a Burundian living in poverty, with a GDP per capita of 233.8 USD in 2020 (The World Bank, 2020). However, since 2005, the government of Burundi has provided free healthcare at public hospitals for pregnant women and children under five, helping to reduce the financial burden on infected women despite frequent reagent shortages (MSPLS, 2021). Unfortunately, medical insurance or government-free healthcare policies may not apply in private hospitals. Medicine shortages in public hospital pharmacies could worsen the situation for women needing spiramycin, forcing them to turn to expensive private pharmacies. In France, a developed country, the total cost was 851 Euros (928 USD) for prenatal screening and 773 Euros (843 USD) for postnatal screening of toxoplasmosis, significantly higher than our estimates (Binqet et al., 2019). The estimated costs did not include the economic burden associated with the postnatal stage, due to a lack of hospital data and follow-up for children with CT sequelae. Otherwise, the overall economic burden was even higher. Awareness campaigns for the Burundian population would be important to reduce the economic and health burden associated with CT. A lack of knowledge about the disease and its transmission to humans was reported (Minani et al., 2025b), which could present a significant barrier to disease prevention. This highlights the need for improved knowledge via health education. The fact that the disease is diagnosed only in pregnant women who visit major hospitals contributes to the limited awareness among the large populations living in rural areas, who rely on their nearest health centres. Therefore, expanding diagnostic tools to all healthcare facilities and ensuring the availability of reagents and drugs for toxoplasmosis testing and treatment is crucial for its control and prevention.

The strength of this study lies in its inclusion of economic impact estimates, which are missing from all other studies on the burden of CT. However, this study had some limitations. The lack of toxoplasmosis diagnostic capacity at health centres serving the largest proportion of the population, IgG titration and avidity testing and test reagent shortages in major hospitals in Burundi would likely have led to uncertainties in seroprevalence based on IgG or IgM antibodies when estimating the incidence of CT. Since none of the hospitals used molecular techniques to diagnose *T. gondii* infection in amniotic fluid samples from pregnant women, the reported prevalence of acute *T. gondii* infection may be biased. However, the prevalence values used in the DALY estimates were generally consistent with estimates reported in some African countries and with global estimates of acute *T. gondii* infection in pregnant women (Rostami et al., 2019). We also employed advanced epidemiological modelling techniques, incorporating appropriate statistical probability distributions to adjust for potential biases in the estimates. In addition, the proportion of seroconversion from mother to foetus was unavailable in Burundi and other African countries. We used 28.1 %, the estimated proportion of seroconversion from mother to foetus based on studies in European countries (Torgerson and Mastroiacovo, 2013), which may not apply to Burundi, a low-income country. We did not specify the sequelae of CT according to genotype categories, which could introduce

bias into our results. In Europe and North America, the sequelae of CT are predominantly associated with Type II genotypes, which tend to cause milder symptoms, whereas in South America, non-Type II genotypes are more common and are linked to more severe sequelae (Torgerson and Mastroiacovo, 2013). Updated disability weights based on the health states were taken into account to obtain accurate DALY estimates (WHO, 2020). The estimated costs related to CT provide only an overview of the economic burden in Burundi; however, undiagnosed pregnant women, lack of follow-up for children, and unreported cases have likely resulted in an underestimation of these costs.

5. Conclusions

This study demonstrates that CT is a significant public health concern in Burundi, although its burden is lower compared to other major infectious diseases in the country. As a foodborne zoonotic disease, control measures such as proper cooking of meat, improved hygiene and sanitation practices, better management of cats and their faeces, routine diagnostic check-ups for pregnant women, and timely treatment for infected pregnant women are essential to reduce the health and economic burden of toxoplasmosis in Burundi. Studies that update the economic and health burden of both congenital and acquired toxoplasmosis, using comprehensive data from all healthcare facilities in Burundi, are needed, as the current study has some uncertainty issues. We recommend routine toxoplasmosis screening in antenatal programs and national investment in diagnostic infrastructure. We urge policy-makers to consider the findings of this study in expanding diagnostic capacity and treatment availability in all healthcare facilities, and to include toxoplasmosis in public health intervention programmes to reduce the incidence of congenital and acquired cases in Burundi.

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Data availability

All relevant data are within the paper and its Supporting Information.

Abbreviations

BIF	Burundi francs
CHUK	Centre Hospitalo Universitaire de Kamenge
CNS	Central nervous system
CT	Congenital toxoplasmosis
DALYs	Disability-adjusted life years
DW	Disability weights
ECDC	European Centre for Disease Prevention and Control
GBD	Global Burden of Disease
GDP	Gross domestic product
HMK	Hôpital Militaire de Kamenge
ICC	Intracranial calcifications
IgG	Immunoglobulin G
IgM	Immunoglobulin M
ISTEEBU	Institut de Statistiques et d'Etudes Economiques du Burundi
MPH	Ministry of Public Health
MSPLS	Ministère de la Santé Publique et de la Lutte contre le Sida
OMS	Organisation Mondiale de la Santé
UI	Uncertainty interval
USD	United States dollars
YLD	Year lived with disability
YLL	Year of life lost
WHO	World Health Organization

CRediT authorship contribution statement

Salvator Minani: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Carlotta Di Bari:** Writing – review & editing, Validation, Methodology. **Brecht Devleesschauwer:** Writing – review & editing, Validation, Methodology. **Anastasio Gasogo:** Writing – review & editing, Supervision, Conceptualization. **Jean-Bosco Ntirandekura:** Writing – review & editing, Supervision, Conceptualization. **Sarah Gabriël:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Chiara Trevisan:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no competing interests.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.actatropica.2025.107828](https://doi.org/10.1016/j.actatropica.2025.107828).

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